An identifiability wall for the fractional order in diffusion-MRI signal decay, and the time/space degeneracy of the joint continuous-time random-walk model

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Abstract

Anomalous-diffusion MRI summarises tissue microstructure through a *fractional order*: the stretched-exponential heterogeneity exponent α, or—in the continuous-time random-walk (CTRW) / fractional Bloch–Torrey picture—a joint time-order α and space-order β. We ask a prior question to estimation: *can the fractional order be recovered at all* from a finite-b, Rician-noise magnitude signal, estimated jointly with the diffusion coefficient D and the amplitude S_0? We derive the Fisher information and Cramér–Rao lower bound (CRLB) for the fractional order under the b-indexed signal-decay forward model and locate a *recovery-collapse wall*. For the single-order stretched-exponential model, under sparse clinical acquisition (4 b-values, $b_{\max} = 2000\,\mathrm{s/mm^2}$) the wall sits at $\mathrm{SNR}^* = 27.8$ (CRLB) / 29.9 (bootstrap, 95 % CI $[28.3, 31.1]$), *inside* the cited clinical-SNR band $[20, 60]$, and it is robust across the physiological α range $[0.60, 0.98]$ (wall in $[27.5, 32.9]$). For the joint model at a single diffusion time, the time- and space-orders are *structurally degenerate*: the median $|\rho_{\alpha\beta}| = 0.984$ and the Fisher matrix is ill-conditioned ($\sim 1.7\times10^8$). Unlike the single-order wall, the degeneracy is *not* relieved by adding b-values; only a second diffusion time separates the orders ($|\rho_{\alpha\beta}| : 0.943 \to 0.182$). Every bound is an *identifiability* statement scoped to its regime (forward model, SNR band, acquisition), never an impossibility claim.

Scope (read first). All results are derived (closed-form Fisher/CRLB + profile-likelihood + parametric bootstrap) on a *fully synthetic* forward model; no in-vivo or clinical data is used. A CRLB is an information lower bound—an *identifiability* statement scoped to a regime, never a claim that a parameter is unknowable in principle. The headline wall is for *sparse clinical* acquisition; it recedes below the clinical band for dense multi-b research acquisition ($n_b \geq 8$).

1 Introduction

Non-Gaussian diffusion in tissue is widely summarised by a fractional order. The stretched-exponential model [1] writes the magnitude signal as $S(b) = S_0 \exp[-(bD)^\alpha]$, with $\alpha \in (0, 1]$ a *heterogeneity index*. The continuous-time random-walk (CTRW) / fractional Bloch–Torrey picture [2] replaces the single exponent by two: a time-fractional (Caputo) order $\alpha \in (0, 1]$ encoding waiting-time heterogeneity and a space-fractional (Riesz) order $\beta \in (1, 2]$ encoding jump-length

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heterogeneity. These orders are reported as biomarkers; the prior question—*whether they are recoverable at all* under realistic acquisition—is rarely priced.

We price it. The estimand is the fractional order, estimated *jointly* with D and S_0 from a finite set of b-values under Rician magnitude noise. The deliverable is a *recovery-collapse wall*: where, as a function of SNR and acquisition, the order ceases to be usefully recoverable, located by the relative CRLB and corroborated by the parameter correlations, the Fisher condition number, and profile-likelihood / parametric-bootstrap confidence intervals.

What this is *not* (positioning). The object is the Fisher information of the fractional order where the parameter enters the likelihood *only through the b-indexed signal attenuation* under Rician magnitude noise. This is a different statistical experiment from three neighbours. (i) The Cramér–Rao bound for the self-similarity / Hurst exponent of fractional Brownian motion [4] is built from the *trajectory / increment / mean-squared-displacement* likelihood of a self-similar *process*: it has no b-value axis, no Rician magnitude likelihood, and no joint D/S_0 nuisance structure. Same word ("fractional exponent"), different likelihood. (ii) Empirical acquisition / quasi-diffusion-imaging design studies [5] tabulate Monte-Carlo estimation error for chosen designs; we instead *derive* the information content in closed form and *locate the wall*, with an explicit $(n_b, b_{\max})$ surface and the (α, β) degeneracy structure. (iii) CRLB-based b-value optimisation for diffusion *kurtosis* [6] bounds a *cumulant* of a Taylor expansion, not a fractional order; the joint time/space-order identifiability question has no kurtosis analogue. This is not an estimator paper—we do not propose a better fitting method; we bound whether the order is recoverable at all.

2 Forward models and the information layer

2.1 Signal-decay forward models

The single-order *lead lane* is the stretched exponential [1]

$$S(b; S_0, D, \alpha) = S_0 \exp\big[-(bD)^{\alpha}\big], \qquad \alpha \in (0, 1]. \tag{1}$$

For the joint model, the CTRW characteristic function [2] is a Mittag-Leffler function E_α of $|q|^\beta\,\Delta^\alpha$. At a single diffusion time Δ, the narrow-pulse relation $b \propto q^2\Delta$ gives $|q|^\beta \propto b^{\beta/2}$, so the signal reduces to

$$S(b; S_0, D, \alpha, \beta) = S_0\, E_\alpha\big[-(bD)^{\beta/2}\big], \tag{2}$$

with $E_\alpha(z) = \sum_{k\geq 0} z^k/\Gamma(\alpha k + 1)$ the one-parameter Mittag-Leffler function. Equation (2) reduces *exactly* to the stretched exponential (1) at $\alpha = 1$ (since $E_1(z) = e^z$), with the heterogeneity exponent equal to $\beta/2$: the single-order model is the time-order-one face of the joint model. With a second diffusion time the attenuation carries the exponent $(\alpha - \beta/2)$ in Δ, which is what allows the two orders to be separated.

We evaluate $E_\alpha(-t)$, $t \geq 0$, $0 < \alpha < 1$, by the Pollard/Mainardi completely-monotone spectral integral [7], validated against the two closed forms $E_1(-t) = e^{-t}$ and $E_{1/2}(-t) = e^{t^2}\mathrm{erfc}(t)$ and an independent adaptive quadrature to $< 5 \times 10^{-6}$; at $\alpha = 1$ the exact exponential is used.

2.2 Fisher information and the wall

Magnitude data are Rician, $\sigma = S_0/\mathrm{SNR}$ at $b = 0$. Because the parameters enter the likelihood only through $\nu_i = S(b_i; \theta)$, the Fisher information matrix is $\mathrm{FIM}(\theta) = \sum_i f(\nu_i/\sigma)\,\sigma^{-2}\,g_i g_i^\top$, where $g_i = \partial S(b_i)/\partial\theta$ and $f(a)$ is the Rician information factor (which tends to the Gaussian limit 1 as

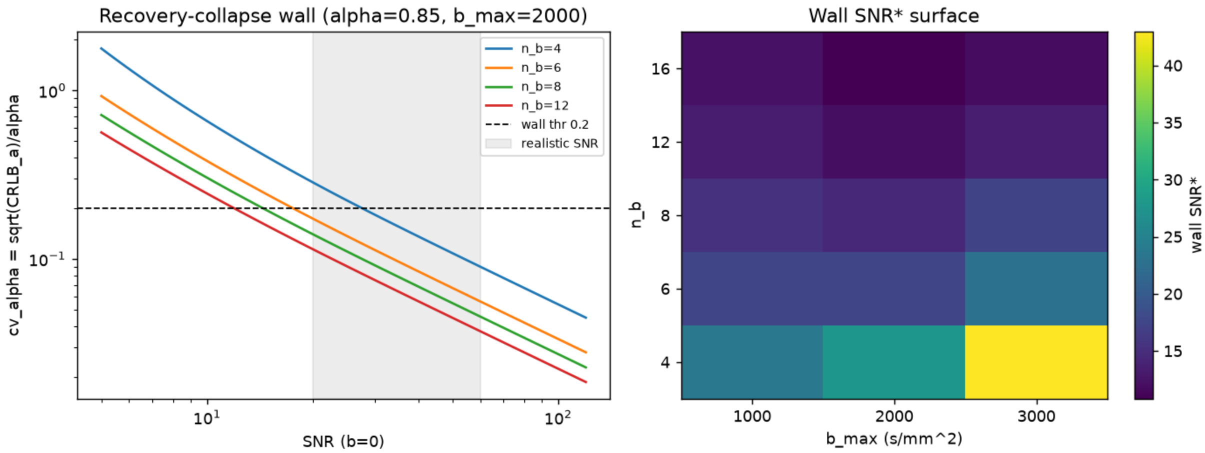

Figure 1: Single-order recovery-collapse wall. *Left:* relative CRLB c_α versus SNR for several b-value counts; the wall is where c_α crosses the pre-registered threshold 0.20 inside the clinical band (shaded). *Right:* wall SNR* over $(n_b, b_{\max})$—n_b dominates.

$a \to \infty$). The CRLB is $\mathrm{cov}(\theta) \succeq \mathrm{FIM}^{-1}$. We locate the single-order wall by the relative CRLB $c_\alpha = \sqrt{[\mathrm{FIM}^{-1}]_{\alpha\alpha}}/\alpha$ crossing a *pre-registered* threshold $c_\alpha = 0.20$, corroborated by the α–D correlation and the FIM condition number, and confirmed by a parametric bootstrap of the joint Rician maximum-likelihood estimator. For the joint model the deliverable is the degeneracy structure of (α, β): the correlation $\rho_{\alpha\beta}$ from FIM^{-1} ($\to \pm 1$ signals inseparability) and the condition number. The fractional order has no elementary Mittag-Leffler order-derivative, so the joint Jacobian is taken by central finite differences (validated against the analytic single-order Jacobian at the $\alpha = 1, \beta = 2$ reduction). All randomness is seeded; the load-bearing intervals reproduce.

3 Results

3.1 The single-order wall (sparse clinical acquisition)

At the canonical sparse clinical protocol ($n_b = 4$, $b_{\max} = 2000\,\mathrm{s/mm^2}$, $\alpha = 0.85$, $D = 1.5 \times 10^{-3}\,\mathrm{mm^2/s}$), the analytic CRLB wall is at $\mathrm{SNR}^* = 27.8$ and the empirical (bootstrap-MLE) wall at $\mathrm{SNR}^* = 29.9$, 95 % CI $[28.3, 31.1]$—both *inside* the cited clinical-SNR band $[20, 60]$ (Section 3.4). Extending $b_{\max}$ with few b-values trades against α–D collinearity ($\rho_{\alpha D} = -0.87$ at $n_b = 4$, $b_{\max} = 3000$, $\mathrm{SNR} = 30$). The number of b-values is the dominant driver: the wall recedes *below* the clinical band only for dense research acquisition ($n_b \geq 8$). Figure 1.

3.2 Robustness across the physiological α range

Sweeping the wall across the physiological stretched-exponential range $[0.60, 0.98]$ at the sparsest clinical acquisition, the wall SNR* stays in $[27.5, 32.9]$—inside the clinical band at *every* α; $\alpha = 0.85$ was not special, and the wall is slightly worse toward low (more heterogeneous) α. The pre-registered refute (the wall receding below the band somewhere across α) did not trigger: the "clinically information-limited" claim is robust across α, not an artefact of a single operating point.

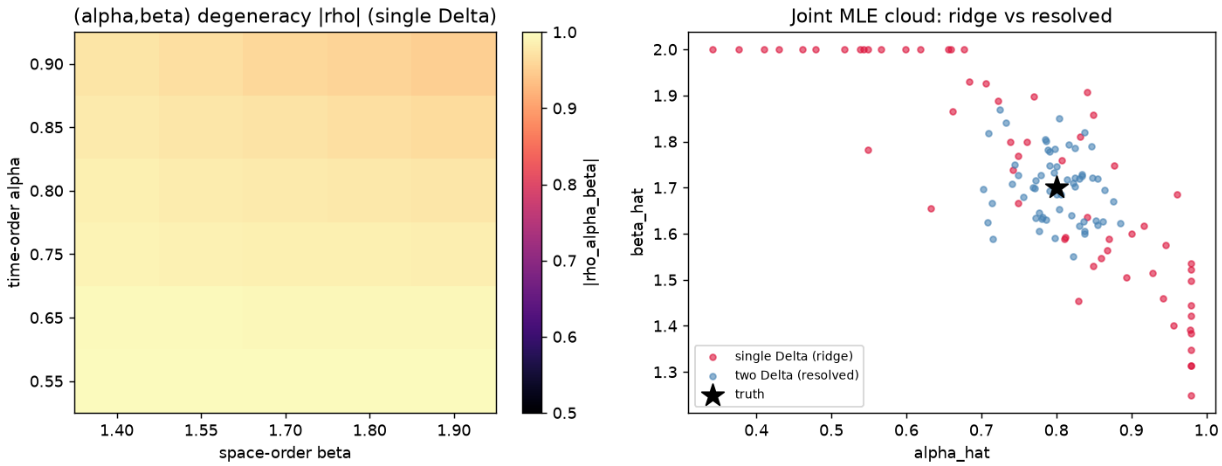

Figure 2: Joint (α, β) degeneracy at a single diffusion time. *Left:* $|\rho_{\alpha\beta}|$ over the physiological grid. *Right:* the joint-MLE cloud collapses onto a ridge at one diffusion time (red) and resolves with a second diffusion time (blue); star is truth.

3.3 The joint time/space degeneracy (single diffusion time)

With both orders free at a single diffusion time, the joint Fisher matrix is severely degenerate across the physiological grid: median $|\rho_{\alpha\beta}| = 0.984$ (range $[0.950, 0.998]$) with condition number $\sim 1.7 \times 10^8$, exceeding the pre-registered degeneracy threshold 0.90. At the headline voxel ($\alpha = 0.80$, $\beta = 1.70$) the analytic $\rho_{\alpha\beta} = -0.979$; the parametric bootstrap (200 replicates) returns $\hat{\alpha}$ 95 % CI $[0.35, 0.98]$ and $\hat{\beta}$ 95 % CI $[1.19, 2.00]$, spanning nearly the entire admissible range—the practical face of the degeneracy.

Crucially, the degeneracy is *not* relieved by adding b-values: $|\rho_{\alpha\beta}|$ moves only from 0.994 at $n_b = 4$ to 0.943 at $n_b = 16$. This is the opposite of the single-order wall (which recedes for $n_b \geq 8$) and is the central distinction of the joint problem: a single decay curve, however densely sampled in b, cannot disentangle a Mittag-Leffler time-order from a b-power space-order. Only a *second diffusion time* breaks it: $|\rho_{\alpha\beta}|$ drops from 0.943 to 0.182 and the relative CRLB c_α from 0.21 to 0.05 (equal total measurement count). Figure 2. This mechanistically explains why historical CTRW characterisations [2] separated the orders only by acquiring more than one experiment family.

3.4 The clinical-SNR band

The band $[20, 60]$ is cited, not asserted: clinical diffusion-weighted imaging reports $b = 0$ SNR of order 40 at 3 T and 70–90 at 7 T [3], placing routine 1.5–3 T acquisition in $[20, 60]$ and research acquisition up to ~ 100. The walls of Sections 3–3.2 sit inside this band; the degeneracy of Section 3.3 holds throughout it.

4 Discussion

Two regime-scoped identifiability statements emerge. First, the single stretched-exponential order is *information-limited* under sparse clinical acquisition—its recovery wall lies inside the clinical-SNR band across the physiological α range—and becomes recoverable only with dense multi-b research acquisition. Second, the joint CTRW time- and space-orders are *structurally degenerate* at a single diffusion time, and—unlike the single order—no amount of b-sampling separates them; a second diffusion time is required. Both are *identifiability* statements scoped to the forward model, SNR band, and acquisition; neither asserts that a parameter is unknowable in principle. The constructive boundaries ($n_b \geq 8$ for the single order; a second diffusion time for the joint orders) are stated, not hidden.

Limitations. The forward models are single-compartment and single-diffusion-time (except where a second time is explicitly added); the noise is Rician with known σ; the analysis is simulation/theory, not an in-vivo study. The bootstrap operates inside box constraints on (α, β), so the empirical $\hat{\alpha}, \hat{\beta}$ correlation understates the analytic $\rho_{\alpha\beta}$—the marginal CIs spanning the admissible range are the honest practical signal. The tooling for the reused Grünwald–Letnikov operators and the noise-amplification cross-check is forward-cited to a companion release [8].

5 Conclusion

Under the b-indexed Rician diffusion-MRI forward model, the stretched-exponential fractional order is information-limited at clinically realistic acquisition, and the joint CTRW time/space orders are degenerate at a single diffusion time in a way that more b-values cannot fix. These are sharp, regime-scoped identifiability limits for fractional-order diffusion biomarkers, and they point to the acquisition change (a second diffusion time) that would lift the joint degeneracy.

Reproducibility. All numbers trace to seeded result files (`results/RESULTS_CP{0,1,2}.json`) regenerated by `reproduce.sh`; every manuscript number is an auto-generated macro checked by `paper/consistency.py`.

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